Engine Drains – General Data

1. General

Refer to Figure 1 for details.

The area surrounding the power plant installation is purged of inflammable vapors by the cooling airflow within the cowlings, but to prevent accumulation of liquids capable of creating a fire hazard, a drain system is provided. The drain system comprises an automatic fuel control (FCU) purge valve, a fuel flow divider/dump valve drain line with check valve, a FCU seepage drain, a starter/generator seal drain and two combustion chamber drains. Waste fuel from the automatic FCU purge valve and fuel flow divider/dump valve is routed through lines and flexible hoses into the fuel system vent line in the wing. Fluid from the FCU seepage drain, starter/generator seal drain and combustion chamber drains is drained overboard through pipes which protrude through the engine lower cowl.

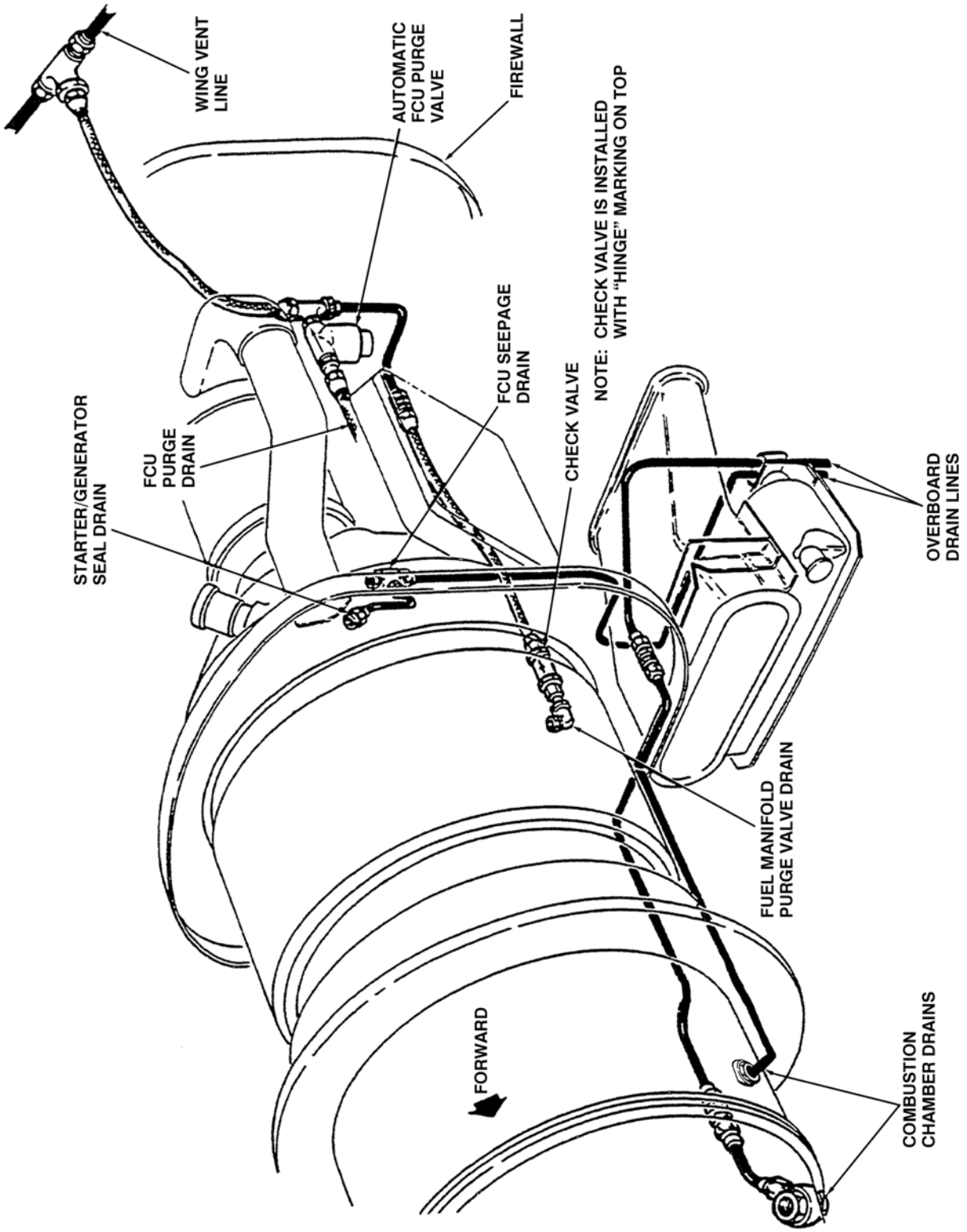
A. Automatic FCU Purge Valve

An automatic FCU purge valve is provided for each engine and is mounted in the engine nacelle aft of the firewall. The valve is energized during the starting cycle to provide a means of purging trapped air from the fuel control unit so that the metering of fuel will not be affected. On completion of the cycle the valve is deenergized (closed).

B. Flow Divider and Dump Valve

The flow divider and dump valve divides the fuel flow from the fuel control unit between the primary and secondary fuel manifolds during engine start-up and operation, and dumps residual fuel from the manifolds at engine shutdown.

When the fuel cut-off valve in the FCU closes during engine shutdown, an operating spring in the flow divider and dump valve overcomes the fuel inlet pressure and moves a piston within the unit body to block the fuel inlet port. This movement of the piston connects both primary and secondary manifolds to the dump valve port and allows residual fuel in the manifold to drain to the fuel system vent in the wing.



Engine Drains
Figure 1

Print Date: 2025-09-18